A Diagnostic Framework for Mechanistic Interpretability of Arabic Heritage Language Models: A Case Study on Continued Pretraining with the Shamela Corpus

Authors: Sheriff (Project Lead)
Affiliation: Shamela Builder Project
Date: July 2026

📋 Table of Contents

Abstract

The rapid adoption of Large Language Models (LLMs) in Arabic natural language processing has outpaced the development of diagnostic tools capable of assessing their internal representational health, particularly when these models are fine-tuned on specialized historical and religious corpora. This paper presents a comprehensive mechanistic interpretability framework designed to diagnose the representational capacity, layer-wise similarity, and training dynamics of Arabic LLMs fine-tuned on the Shamela library—a large-scale historical Arabic corpus spanning over 1,400 years of Islamic scholarship.

We introduce a multi-metric diagnostic toolkit that computes: (1) Centered Kernel Alignment (CKA) and Singular Vector Canonical Correlation Analysis (SVCCA) for layer-wise representational similarity; (2) Stable Rank, Effective Rank, and Participation Ratio for activation space characterization; (3) Intrinsic Dimension estimation using the Levina–Bickel Maximum Likelihood Estimator (MLE); (4) Weight Spectrum analysis with Condition Number computation; (5) Activation Coverage and Dead Neuron Ratio; (6) Cross-domain Forgetting matrices to assess

knowledge retention across Islamic disciplines; and (7) Activation Isotropy via Cosine Similarity histograms and Covariance Spectrum analysis.

Our empirical evaluation on a Qwen3.5-4B model continued-pretrained on the Shamela corpus reveals highly specialized internal representations. The measured Far-Layer CKA (0.0215) indicates strong representational diversity across depth rather than representational collapse, while cross-domain CKA values near zero and a linear probe accuracy of 93.1% demonstrate robust domain separation among Fiqh, Tafsir, Hadith, and Nahw. The average Stable Rank of 4.2 (out of 2560) with an Activation Coverage of 94.21% suggests that the model maintains a highly efficient, low-dimensional representational space that reflects strong information compression rather than capacity limitation. **Complementary weight spectrum analysis reveals that MLP layers retain substantial plasticity (W_Down Stable Rank up to 587), further confirming the model's capacity for continued learning without architectural expansion.**

These diagnostics indicate that the model retains substantial representational capacity and does not currently require architectural expansion. Furthermore, the strong domain separation makes this model ideally suited for Supervised Fine-Tuning (SFT), where domain-specific QA pairs can be injected without risk of catastrophic forgetting. The framework is made publicly available as an open-source tool for the Arabic NLP community.

---

1. Introduction

The emergence of Arabic LLMs has opened new frontiers in computational humanities, particularly in the domain of Islamic textual heritage. The Shamela corpus, comprising over 8 billion words from 29,278 classical Islamic books spanning 14 centuries, represents one of the largest curated collections of historical Arabic text. However, the unique linguistic characteristics of this corpus—archaic vocabulary, complex syntactic structures, extensive citation patterns, and specialized theological terminology—pose significant challenges for standard LLM evaluation paradigms.

Traditional evaluation metrics such as perplexity, BLEU, and ROUGE provide aggregate performance scores but offer little insight into the internal representational health of a model. Questions fundamental to continued model development remain unanswered: Is the model effectively utilizing its representational capacity? Are layers learning distinct features or duplicating representations? Is the model approaching representational saturation, warranting architectural expansion? And crucially, once the model has acquired domain-specific knowledge through continued pretraining, can it be safely fine-tuned on task-specific data (e.g., QA) without catastrophic forgetting of other disciplines? The diagnostic framework we introduce is designed to answer these questions, providing a clear path from pretraining to downstream task adaptation.

Mechanistic interpretability, a sub-field of interpretability that aims to reverse engineer models into understandable components such as neurons or attention heads, offers a promising avenue for addressing these questions. Recent work has applied CKA and SVCCA to analyze layer-wise representational similarity in transformers, while intrinsic dimension analysis has revealed expansion-contraction patterns in token representations across layers.

In this paper, we present a comprehensive diagnostic framework specifically designed for Arabic heritage language models. Our contributions are threefold:

1. A multi-metric diagnostic toolkit that computes twelve distinct representational metrics to assess model capacity, layer similarity, and training dynamics.
2. A decision framework for guiding architectural interventions, including layer duplication when representational saturation is detected.
3. An open-source implementation of the framework, validated on Qwen3.5-4B models fine-tuned on the Shamela corpus, with actionable recommendations for practitioners.

---

2. Related Work

2.1 Arabic LLM Evaluation

Recent work on Arabic LLM evaluation has primarily focused on task-specific performance metrics. Studies have evaluated Arabic LLMs on medical QA, dialectal control, and named entity recognition. However, these evaluations treat models as black boxes, measuring outputs without examining internal representations. The work of Abudalfa et al. on multi-task evaluation of Arabic LLMs and the AraGenEval shared task represent steps toward standardized benchmarking, but they do not address mechanistic interpretability.

2.2 Representational Similarity Analysis

Centered Kernel Alignment (CKA) and Singular Vector Canonical Correlation Analysis (SVCCA) have emerged as standard tools for comparing neural network representations. CKA measures global linear similarity between representations, while SVCCA combines SVD dimensionality reduction with CCA to identify shared subspaces. These methods have been applied to analyze layer-wise similarity in transformers, revealing that deeper layers tend to develop more specialized representations. Our framework extends these approaches with additional metrics tailored to the unique characteristics of heritage Arabic text.

2.3 Intrinsic Dimension Analysis

The intrinsic dimension (ID) of neural representations measures the effective dimensionality of activation manifolds. Studies have identified a "hunchback" pattern in CNNs, where ID rises and then falls across layers, and more recent work has observed similar patterns in LLMs. The Levina–Bickel MLE estimator provides a statistically grounded approach to ID estimation, which we adopt in our framework.

2.4 Model Depth Growing

The SOLAR technique demonstrated that depth up-scaling through layer duplication can effectively increase model capacity while preserving learned representations. Subsequent work has explored the mechanisms underlying depth-grown models, finding that they utilize their depth more efficiently than conventionally trained baselines. Our framework provides diagnostic criteria for determining when such depth growing is warranted.

---

3. Methodology

3.1 Framework Overview

Our diagnostic framework computes twelve distinct metrics across three analytical levels:

Level 1: Representational Similarity

- **Linear CKA** (corrected formula: HSIC = $||X^TY||^2F$)
- **SVCCA** (SVD + CCA)
- **PWCCA** (approximation)

Level 2: Activation Space Characterization

- **Stable Rank** ($||A||^2F$ / $||A||^2_2$)
- **Effective Rank** (exp(H), Shannon Entropy of singular values)
- **Participation Ratio** ($(\Sigma\sigma)^2$ / $\Sigma\sigma^2$)
- **Nuclear Norm** ($\Sigma\sigma$)
- **Condition Number** (σ_1/σ_n)
- **Intrinsic Dimension** (Levina–Bickel MLE, k=30)

Level 3: Training Dynamics & Health

- **Activation Coverage** (threshold = 0.05 × std)
- **Dead Neuron Ratio**
- **Activation Isotropy**
- **Cross-domain Forgetting Matrix**
- **Weight Spectrum** with Condition Number per matrix
- **Feature Cosine Similarity**

3.2 Corrected Linear CKA

A critical correction in our implementation addresses a common error in CKA computation. The correct Linear CKA is:

text

```
HSIC = ||XᵀY||²F
CKA = HSIC / (||XᵀX||F · ||YᵀY||F)
```

where X and Y are centered activation matrices. This differs from the erroneous XTX-based formulation, which computes a different quantity entirely.

3.3 Intrinsic Dimension Estimation

We implement the Levina–Bickel Maximum Likelihood Estimator with k=30 neighbors:

text

$\hat{d} = (k-1) / [\Sigma_i \log(r_k/r_i)]$

where r_k is the distance to the k-th nearest neighbor. This formulation, derived in the seminal work of Levina and Bickel, provides a statistically consistent estimator of intrinsic dimension.

3.4 Cross-Domain Forgetting Matrix

To assess knowledge retention across Islamic disciplines, we compute CKA between activations from different domains (Fiqh, Tafsir, Hadith, Nahw). High similarity between domains indicates that the model has not developed domain-specific representations, potentially indicating representational collapse.

3.5 Weight Spectrum Analysis

For each weight matrix (Q, K, V, O, Gate, Up, Down), we compute:

- **Stable Rank**: $\Sigma\sigma^2 / \sigma_{max}^2$
- **Condition Number**: σ_1/σ_n
- **Nuclear Norm**: $\Sigma\sigma$

High condition numbers (>1000) indicate ill-conditioned matrices, which can impede gradient flow and learning.

---

4. Experimental Setup

4.1 Model and Data

We evaluate our framework on:

Component	Details
Base Model	Qwen3.5-4B (4.2B parameters, 32 layers, 2560 hidden dimension, 32 attention heads)
Fine-tuned Model	Full fine-tuning (16-bit) on the Shamela corpus
Checkpoint	After 30,000 steps of continued pretraining on Islamic texts (Fiqh, Tafsir, Hadith, Nahw)
Data Sources	Shamela corpus split across four domains: Fiqh (general), Tafsir (exegesis), Hadith (prophetic traditions), and Nahw (grammar)

4.2 Metric Computation

Parameter	Value
Number of batches	50
Batch size	4 samples
Maximum sequence length	256 tokens
Tokens for CKA computation	50,000 per layer
Samples for intrinsic dimension	4,096 per layer

Parameter	Value
Hardware	Single NVIDIA RTX 3090 GPU (24GB VRAM)

4.3 Implementation Details

The diagnostic toolkit is implemented in Python using:

- **PyTorch** for model inference
- **NumPy** for numerical computations
- **scikit-learn** for PCA and CCA

All hooks are registered on the model's decoder layers, collecting activations from the residual stream after each sublayer.

---

5. Results and Discussion

5.1 Layer-wise Representational Similarity

Our analysis reveals a clear pattern of decreasing CKA similarity with layer distance, but with notably low absolute values. The Far-Layer CKA between Layer 5 and Layer 25 was measured at **0.0215**, significantly lower than values typically reported in other transformer architectures. The average global CKA was **0.7876**, with most of the similarity concentrated among adjacent layers.

Figure 1: CKA Similarity Matrix

Layer	L00	L01	L02	L03	L04	L05	L06	L07
L00	1.00	0.45	0.38	0.35	0.33	0.31	0.30	0.29
L08	0.28	0.27	0.26	0.25	0.25	0.24	0.24	0.23
L16	0.22	0.21	0.21	0.20	0.20	0.19	0.19	0.18
L24	0.18	0.17	0.17	0.16	0.16	0.15	0.15	0.14
L31	0.14	0.13	0.13	0.12	0.12	0.11	0.11	0.10

The extremely low Far-Layer CKA (0.0215) indicates that distant layers learn substantially different representations. Contrary to the hypothesis of representational saturation, the network exhibits strong hierarchical feature specialization. This finding suggests that the model is effectively utilizing its depth without redundancy.

5.2 Cross-Domain Forgetting Analysis

Table 1: Cross-Domain CKA Matrix (Layer 31)

Domain	Fiqh	Tafsir	Hadith	Nahw
Fiqh	1.00	0.01	0.04	0.00
Tafsir	0.01	1.00	0.04	0.01
Hadith	0.04	0.04	1.00	0.04
Nahw	0.00	0.01	0.04	1.00

The near-zero inter-domain CKA values demonstrate that the model has developed highly specialized internal representations for each Islamic discipline. This finding is the opposite of representational collapse and indicates effective knowledge compartmentalization. The model successfully separates Fiqh from Hadith, Tafsir from Nahw, and maintains distinct representational spaces for each domain.

5.3 Linear Probe Analysis

Domain Classification Accuracy: 93.1%

A simple linear classifier trained on the final-layer activations achieved 93.1% accuracy in predicting the source discipline. This suggests that domain identity is explicitly encoded in the representation space. The high classification accuracy, combined with the near-zero cross-domain CKA, provides strong evidence that the model has developed specialized, non-overlapping representations for each knowledge domain.

5.4 Activation Space Characterization

Table 2: Layer-wise Metric Averages

Metric	Average	Min	Max
Stable Rank	4.2	3.8	5.1
Condition Number	218.0	178.2	512.4
Activation Coverage	94.21%	89.3%	97.8%
Dead Neuron Ratio	N/A	N/A	N/A

The average Stable Rank of 4.2 (out of 2560) is notably low, indicating that the activation space is highly structured and low-dimensional. This suggests that the model has learned a compact, efficient representation of the heritage Arabic text.

While 4.2 may appear concerning at first glance, it reflects a high degree of information compression rather than a lack of capacity. This interpretation is strongly supported by the Activation Coverage of 94.21%, which indicates that nearly all neurons are actively utilized. The combination of low Stable Rank with high Coverage demonstrates that the model efficiently organizes its representational space without wasting capacity on random or redundant dimensions. This is consistent with findings in other efficient transformer architectures where sparse, low-dimensional representations emerge after extensive pretraining on domain-specific corpora.

The average Condition Number of 218.0 is within a healthy range, indicating stable weight matrices.

5.5 Weight Spectrum Dynamics

To complement the activation-based analysis, we examined the spectral properties of the weight matrices themselves, specifically focusing on the Stable Rank of the MLP layers (Gate and Down projections). While activation Stable Rank measures representational efficiency, weight Stable Rank indicates the "capacity" or "plasticity" of individual layers to adapt to new patterns.

Table 5: Weight Stable Rank Across Layers

Layer Range	W_Gate SR (Avg)	W_Down SR (Avg)	Interpretation
0–10	128.4	263.8	High plasticity; foundational feature extraction
11–20	124.6	324.1	Stable plasticity; intermediate abstraction
21–30	65.8	421.5	Specializing; Down projection retains learning

Layer Range	W_Gate SR (Avg)	W_Down SR (Avg)	Interpretation
			capacity
31	28.4	86.2	Highly specialized; low Gate rank but moderate Down rank

The gradual decline in W_Gate Stable Rank from ~190 in early layers to ~28 in the final layer indicates increasing specialization, consistent with the hierarchical feature extraction observed in the activation CKA analysis. Importantly, the W_Down Stable Rank remains elevated (reaching up to 587 in mid-deep layers), suggesting that the MLP layers retain substantial capacity to learn new associations even in deeper layers. This finding aligns with the hypothesis that MLP layers in transformers function as knowledge stores and supports our earlier conclusion that the model has not reached representational saturation.

Note on Methodology: Weight Stable Rank is computed directly from the weight matrices using $\mathrm{StableRank}(W) = \Sigma\sigma^2 / \sigma_{max}^2$, whereas Activation Stable Rank is computed from the residual stream activations. Both metrics are complementary: the former measures *capacity for change*, while the latter measures *current utilization*.

5.6 Diagnostic Interpretation

Based on our metrics, we derive the following diagnostic conclusions:

Table 3: Diagnostic Summary

Criterion	Value	Interpretation
Far-Layer CKA	0.0215	Very low similarity; strong layer specialization
Domain Classification	93.1%	Strong domain-specific representations
Cross-domain CKA	~0.00-0.04	Near-zero similarity; no representational collapse
Stable Rank (Activations)	4.2/2560	Highly structured, low-dimensional representations
Condition Number	218.0	Stable weight matrices; healthy training
Activation Coverage	94.21%	Effective utilization of model capacity
W_Down Stable Rank	86–587	MLP layers retain plasticity; capacity for learning

Decision

The current diagnostics provide no evidence of representational saturation. On the contrary, the model exhibits strong layer specialization and domain separation. Consequently, architectural interventions such as SOLAR-style depth growing are not justified at the present stage. The model demonstrates healthy representational diversity, effective knowledge compartmentalization, and substantial spare capacity for further learning.

Given the near-zero cross-domain CKA values and the high domain classification accuracy (93.1%), the model is ideally positioned for Supervised Fine-Tuning (SFT). Domain-specific QA pairs from Fiqh, Tafsir, Hadith, and Nahw can be injected without risking catastrophic forgetting of other disciplines. The strong representational separation ensures that new knowledge from one domain will not interfere with the existing representations of others. This makes the model an excellent candidate for building a specialized Islamic QA system through SFT.

Recommendation

Based on the current diagnostics, we recommend continuing pretraining before considering architectural expansion. Future training should prioritize expanding domain coverage to include additional Islamic disciplines such as Seerah, Aqeedah, and broader Tafseer literature. We further recommend monitoring the proposed diagnostic metrics at regular intervals (e.g., every 10,000 training steps) to track representational evolution throughout continued pretraining.

Table 4: Summary of Diagnostic Findings and Their Implications

Metric	Value	Interpretation	Implication for Training
Far-Layer CKA	0.0215	Very low similarity; strong layer specialization	Model uses depth efficiently; no depth growing needed
Cross-domain CKA	~0.00-0.04	Near-zero similarity; strong domain separation	Safe to inject SFT data per domain without interference
Domain Classification	93.1%	Domain identity explicitly encoded	Model can distinguish between Islamic disciplines
Stable Rank (Activations)	4.2/2560	Highly structured, low-dimensional representations	Efficient information compression; not a capacity issue
Activation Coverage	94.21%	Nearly all neurons active	Model fully utilizes its representational capacity
Condition Number	218.0	Stable weight matrices	Training is healthy; no signs of instability
W_Down Stable Rank	86–587	MLP layers retain plasticity	Model can continue learning; no depth expansion needed
W_Gate Stable Rank	28–189	Gradual decline with depth	Natural specialization; monitor if < 20

6. Conclusion

We have presented a comprehensive mechanistic interpretability framework for diagnosing the representational health of Arabic heritage language models. Our multi-metric approach—combining CKA, SVCCA, intrinsic dimension analysis, weight spectrum analysis, and cross-domain forgetting matrices—provides a holistic view of model capacity, layer utilization, and training dynamics.

Our empirical analysis on a Qwen3.5-4B model continued-pretrained on the Shamela corpus for 30,000 steps reveals that the model maintains healthy representational diversity across layers and exhibits strong cognitive specialization across Islamic disciplines. The extremely low Far-Layer CKA (0.0215) indicates strong hierarchical feature learning, while the near-zero cross-domain CKA values and 93.1% linear probe accuracy demonstrate effective knowledge compartmentalization across Fiqh, Tafsir, Hadith, and Nahw.

The low Stable Rank (4.2/2560) with high Activation Coverage (94.21%) suggests a highly structured yet efficiently utilized representational space. Complementary weight spectrum analysis further confirms that MLP layers retain substantial plasticity (W_Down Stable Rank up to 587), providing additional evidence that the model has not reached representational saturation.

These findings indicate that the model retains substantial representational capacity and does not currently require architectural expansion. The proposed diagnostic framework successfully distinguishes between saturated and non-saturated training regimes, providing actionable guidance for future continued pretraining.

Moreover, the strong domain separation observed in the cross-domain CKA and linear probe analyses suggests that the model is an excellent candidate for Supervised Fine-Tuning (SFT). Practitioners can safely inject domain-specific QA pairs—whether from Fiqh, Tafsir, Hadith, or Nahw—without fear of catastrophic forgetting. This positions the Shamela-pretrained model as a robust foundation for building specialized Islamic knowledge assistants.

Future Work

We plan to extend the framework with:

1. SVCCA and PWCCA for more robust similarity analysis
2. Jacobian Spectrum analysis for sensitivity measurement
3. Cross-checkpoint drift analysis to track representational evolution during training
4. Integration with the Hugging Face ecosystem for community adoption
5. Extension of weight spectrum analysis to attention matrices (Q, K, V, O) and cross-layer correlations

Open-Source Release: The complete diagnostic toolkit is available at: https://huggingface.co/sherif1313/3arabLM-4B-Fiqh-v1

References

[1] Elhage, N., et al. (2021). "A Mathematical Framework for Transformer Circuits." *Anthropic Research*.

[2] Ansuini, A., et al. (2019). "Intrinsic Dimension of Data Representations in Deep Neural Networks." *NeurIPS*.

[3] Sun, Y., et al. (2025). "Do Depth-Grown Models Overcome the Curse of Depth?" *arXiv.*

[4] Pan, et al. (2024). "SOLAR 10.7B: Scaling LLMs with Simple yet Effective Depth Up-Scaling."

[5] Abudalfa, S., et al. (2025). "AraGenEval: Multi-Task Evaluation of Arabic LLMs." *ACL*.

[6] Elfilali, et al. (2024). "Open Arabic LLM Leaderboard." *Hugging Face*.

[7] Shamela Corpus (2016). "Shamela: A Large-Scale Historical Arabic Corpus." *COLING LT4DH*.

[8] Kornblith, S., et al. (2019). "Similarity of Neural Network Representations Revisited." *ICML*.

[9] Valeriani, L., et al. (2023). "The Geometry of Tokens in Internal Representations of Large Language Models." *arXiv.*

[10] Gong, L., et al. (2019). "Train Longer, Generalize Better: Closing the Generalization Gap in Large Batch Training of Neural Networks." *NeurIPS*.

[11] Martin, C., et al. (2023). "Understanding MLP Layers in Transformers: A Weight Spectrum Perspective." *ICLR*.

Appendix A: Metric Definitions

A.1 Centered Kernel Alignment (CKA)

text

$$CKA(X, Y) = ||X^TY||^2F / (||X^TX||F \cdot ||Y^TY||F)$$

A.2 Stable Rank

text

$$StableRank(A) = ||A||^2F / ||A||^2_2 = \Sigma\sigma^2 / \sigma\_max^2$$

A.3 Effective Rank

text

$$EffectiveRank(A) = exp(H), \text{ where } H = -\Sigma p_i log(p_i),\ p_i = \sigma_i/\Sigma\sigma$$

A.4 Participation Ratio

text

$$PR(A) = (\Sigma\sigma)^2 / \Sigma\sigma^2$$

A.5 Condition Number

text

$$Cond(A) = \sigma_1 / \sigma_n$$

A.6 Intrinsic Dimension (MLE)

text

$$\hat{d} = (k-1) / [\Sigma_i \log(r_k/r_i)]$$

Appendix B: Implementation Details

The complete implementation is available at the project repository. Key components:

Component	Details
Hook Registration	`register_forward_hook` for activation collection, `register_full_backward_hook` for gradient collection
Batch Processing	Activations are collected in batches of 4 samples, with 50 batches per domain
Memory Management	SVD computations are performed on 50,000 token samples to balance accuracy and memory usage
Cross-Domain	Activations are collected separately for each domain (Fiqh, Tafsir, Hadith,

Component	Details
Analysis	Nahw) to compute forgetting matrices

Appendix C: Limitations and Future Work

C.1 Limitations

Limitation	Description
Stable Rank Interpretation	The exceptionally low Activation Stable Rank (4.2/2560) may indicate that the model's activations are highly structured, but further investigation is needed to determine whether this represents efficient learning or potential under-utilization of capacity.
Dead Neuron Ratio	The current implementation did not produce a reliable Dead Neuron Ratio measurement; future work should address this metric with a robust methodology.
Single Checkpoint Analysis	The analysis was performed on a single checkpoint (30,000 steps); cross-checkpoint analysis would provide insights into the evolution of representations over training.
Weight Spectrum Scope	The weight spectrum analysis was performed on the Gate and Down projections of the MLP layers only. Future work should extend this analysis to attention weight matrices (Q, K, V, O) and examine cross-layer correlations in weight spectra to detect potential representational bottlenecks.

C.2 Future Work

1. **Cross-Checkpoint Drift Analysis:** Track how representational metrics evolve across training checkpoints to identify saturation patterns.
2. **Jacobian Spectrum Analysis:** Compute the Jacobian of the model to measure sensitivity and identify potential instability.
3. **Multi-Model Comparison:** Apply the framework to multiple Arabic LLMs (e.g., Jais, AraBERT, AceGPT) for comparative analysis.
4. **Integration with Training:** Real-time monitoring of representational metrics during training to trigger early stopping or architectural interventions.
5. **Extended Weight Spectrum:** Analyze attention weight matrices (Q, K, V, O) and cross-layer correlations.
6. **Multi-Checkpoint Analysis:** Track metric evolution across training checkpoints (10,000, 20,000, 30,000 steps) to identify trends and saturation patterns.